

Zaman Serisi Sınıflandırma Pipeline

Kapsamlı Sonuç Raporu

Metrik	Değer
Pipeline katmanları	Gate → Base → Single/Multi Anom → Meta
Toplam model	16 (1 gate + 3 base + 5 single + 5 multi + meta)
Sınıf taksonomisi	34 grup (G01–G34)
Sentetik veri — FULL doğruluk	70.6% (480/680)
Gerçek veri — FULL	11/39 (28.2%)
Gerçek veri — PARTIAL	23/39
Gerçek veri — NONE	5/39

Tüm Modeller — Test Sonuçları

Renk: yeşil ≥ 0.93 | sarı ≥ 0.85 | kırmızı < 0.85

Model	Katman	Kazanan	Test F1	Test Acc
stationarity_gate	Gate	XGBoost	0.9728	0.9724
deterministic trend	Base	LightGBM	0.8620	0.8561
stochastic trend	Base	LightGBM	0.8962	0.8965
volatility	Base	LightGBM	0.8227	0.8182
collective anomaly	Single-Anom	LightGBM	0.9925	0.9925
mean shift	Single-Anom	LightGBM	0.8960	0.8950
point anomaly	Single-Anom	LightGBM	1.0000	1.0000
trend shift	Single-Anom	XGBoost	0.9487	0.9475
variance shift	Single-Anom	LightGBM	0.9554	0.9550
collective anomaly	Multi-Anom	LightGBM	0.9950	0.9950
mean shift	Multi-Anom	LightGBM	0.9333	0.9325
point anomaly	Multi-Anom	LightGBM	0.9899	0.9900
trend shift	Multi-Anom	LightGBM	0.9258	0.9275
variance shift	Multi-Anom	XGBoost	0.9620	0.9625
collective anomaly	Meta	blend	0.9843	0.9926
mean shift	Meta	blend	0.7782	0.9026
point anomaly	Meta	blend	0.9881	0.9945
trend shift	Meta	blend	0.8923	0.9485
variance shift	Meta	blend	0.9344	0.9688

1.1 Gate

Model	Kazanan	LGB	XGB	MLP	Test F1	Test Acc	Pos	Neg
stationarity_gate	XGBoost	0.9557	0.9652	0.9268	0.9728	0.9724	1000	990

1.2 Base Modeller

Model	Kazanan	LGB Val F1	XGB Val F1	MLP Val F1	Test F1	Test Acc	Pos	Neg
deterministic trend	LightGBM	0.9091	0.9082	0.8283	0.8620	0.8561	996	980
stochastic trend	LightGBM	0.8660	0.8586	0.8085	0.8962	0.8965	996	980
volatility	LightGBM	0.7905	0.7810	0.7238	0.8227	0.8182	996	980

1.3 Single-Anomaly Modeller

Anomali	Kazanan	LGB Val F1	XGB Val F1	MLP Val F1	Test F1	Test Acc
collective anomaly	LightGBM	0.9746	0.9691	0.8660	0.9925	0.9925
mean shift	LightGBM	0.9074	0.8940	0.8259	0.8960	0.8950
point anomaly	LightGBM	0.9950	0.9901	0.9659	1.0000	1.0000
trend shift	XGBoost	0.9406	0.9600	0.8911	0.9487	0.9475
variance shift	LightGBM	0.9353	0.9171	0.7879	0.9554	0.9550

1.4 Multi-Anomaly Modeller

Anomali	Kazanan	LGB Val F1	XGB Val F1	MLP Val F1	Test F1	Test Acc
collective anomaly	LightGBM	0.9950	0.9950	0.9394	0.9950	0.9950
mean shift	LightGBM	0.9469	0.9327	0.7900	0.9333	0.9325
point anomaly	LightGBM	0.9851	0.9802	0.9557	0.9899	0.9900
trend shift	LightGBM	0.9543	0.9442	0.9073	0.9258	0.9275
variance shift	XGBoost	0.9796	0.9950	0.9000	0.9620	0.9625

2. 34-Sınıf Sentetik Veri Doğrulama

FULL: 480/680 (70.6%) | PARTIAL: 160 | NONE: 40

GID	Grup Adı	FULL	PARTIAL	NONE	Toplam	Başarı %	Durum
G01	stationary	20	0	0	20	100%	☐ İyi
G02	det trend	17	3	0	20	85%	☐ İyi
G03	stoch trend	13	7	0	20	65%	~ Orta
G04	volatility	8	2	10	20	40%	☐ Zayıf
G05	anom only collective	17	3	0	20	85%	☐ İyi
G06	anom only mean	13	4	3	20	65%	~ Orta
G07	anom only point	17	3	0	20	85%	☐ İyi
G08	anom only trend shift	9	6	5	20	45%	☐ Zayıf
G09	anom only variance	9	9	2	20	45%	☐ Zayıf
G10	det collective	19	1	0	20	95%	☐ İyi
G11	det mean	10	9	1	20	50%	☐ Zayıf
G12	det point	15	5	0	20	75%	~ Orta
G13	det trend shift	7	11	2	20	35%	☐ Zayıf
G14	det variance	14	6	0	20	70%	~ Orta
G15	stoch collective	17	3	0	20	85%	☐ İyi
G16	stoch mean	13	6	1	20	65%	~ Orta
G17	stoch point	12	8	0	20	60%	~ Orta
G18	stoch trend shift	7	11	2	20	35%	☐ Zayıf
G19	stoch variance	11	8	1	20	55%	☐ Zayıf
G20	vol collective	15	5	0	20	75%	~ Orta
G21	vol mean	10	9	1	20	50%	☐ Zayıf
G22	vol point	13	7	0	20	65%	~ Orta
G23	vol trend shift	9	9	2	20	45%	☐ Zayıf
G24	vol variance	15	5	0	20	75%	~ Orta
G25	multi col mean	16	2	2	20	80%	☐ İyi
G26	multi col point	18	2	0	20	90%	☐ İyi
G27	multi mean variance	15	3	2	20	75%	~ Orta
G28	multi point variance	18	0	2	20	90%	☐ İyi
G29	multi point trend	16	3	1	20	80%	☐ İyi
G30	multi col trend	16	2	2	20	80%	☐ İyi
G31	multi col variance	20	0	0	20	100%	☐ İyi
G32	multi mean point	18	2	0	20	90%	☐ İyi
G33	multi mean trend	4	9	7	20	20%	☐ Zayıf
G34	multi trend variance	16	4	0	20	80%	☐ İyi

FULL = base + tüm anomaliler doğru | PARTIAL = kısmen doğru | NONE = hiç doğru değil | Renk: yeşil ≥ 80% | sarı ≥ 60% | kırmızı < 60%

3. Gerçek Veri Değerlendirmesi — 39 Seri

FULL: 11/39 (28.2%) | PARTIAL: 23 | NONE: 5

Etiket	Adet	Oran
FULL — base + tüm anomaliler doğru	11	28.2%
PARTIAL — kısmen doğru	23	59.0%
NONE — hiç doğru değil	5	12.8%

Seri Bazında Tahmin Karşılaştırması

CSV	n	GT Base	GT Anom	Pred Base	Pred Anom	Yol	Match	P_ga te	p_c omb o
airpass	144	stochastic trend	—	stochastic trend	variance shift	combo	PARTIAL	0.000 5	0.93 44
deaths	72	stochastic trend	—	stationary	—	stat_gate	NONE	0.798 3	0.04 00
German_consumption	93	stochastic trend	—	stochastic trend	—	single	FULL	0.001 1	0.11 30
GermanGNP	88	deterministic trend	trend shift	stochastic trend	—	single	NONE	0.001 4	0.04 75
INDPRO	372	stochastic trend	—	stochastic trend	—	single	FULL	0.000 6	0.08 92
np_cpi	129	stochastic trend	trend shift	stochastic trend	—	single	PARTIAL	0.000 5	0.17 55
np_employmt	99	stochastic trend	trend shift	stochastic trend	—	single	PARTIAL	0.001 4	0.02 51
np_gnpdefl	100	stochastic trend	trend shift	stochastic trend	—	single	PARTIAL	0.001 2	0.00 43
np_gnpperca	80	stochastic trend	trend shift	stochastic trend	—	single	PARTIAL	0.000 8	0.24 97
np_indprod	129	stochastic trend	trend shift	stochastic trend	—	single	PARTIAL	0.000 6	0.03 10
np_interest	89	stochastic trend	—	stochastic trend	variance shift	combo	PARTIAL	0.000 7	0.95 42
np_M	100	stochastic trend	trend shift	stochastic trend	—	single	PARTIAL	0.001 5	0.02 57
np_nomgnp	80	stochastic trend	trend shift	stochastic trend	—	single	PARTIAL	0.001 0	0.01 12
np_realgnp	80	stochastic trend	trend shift	stochastic trend	—	single	PARTIAL	0.000 9	0.02 89
np_realwag	89	stochastic trend	trend shift	stochastic trend	—	combo	PARTIAL	0.001 2	0.35 19
np_sp500	118	stochastic trend	trend shift	stochastic trend	—	single	PARTIAL	0.001 0	0.01 79
np_unemploy	99	stationary	—	stationary	—	stat_gate	FULL	0.122 5	0.77 36
np_velocity	120	stochastic trend	—	stochastic trend	—	combo	FULL	0.001 4	0.98 57
np_wages	89	stochastic trend	trend shift	stochastic trend	—	single	PARTIAL	0.001 2	0.03 67
Polish_productivity	117	stochastic trend	trend shift	stochastic trend	—	single	PARTIAL	0.005 7	0.02 09

RealInt_dataframe	103	stationary	mean shift	stationary	trend shift, variance shift	combo_trick	PARTIAL	0.000 6	1.00 00
rec_dataframe	453	stationary	—	stationary	—	stat_gate	FULL	0.257 2	0.54 50
soi_dataframe	453	stationary	—	stationary	—	stat_gate	FULL	0.996 8	0.99 76
strikes	30	stationary	—	stationary	—	stat_gate	FULL	0.254 6	0.98 97
sunspots	100	stationary	—	stationary	—	stat_gate	FULL	0.550 2	0.16 37
tcpd_debt_ireland	21	deterministic trend	trend shift	stationary	—	stat_gate	NONE	0.101 2	0.76 18
tcpd_lga_passengers	468	stochastic trend	mean shift	stochastic trend	—	combo	PARTIAL	0.007 7	0.30 55
tcpd_nile	100	stationary	mean shift	stationary	—	stat_gate	PARTIAL	0.305 9	0.99 97
tcpd_ozone	54	stochastic trend	trend shift	stochastic trend	—	combo	PARTIAL	0.001 9	0.95 73
tcpd_robocalls	52	stochastic trend	mean shift	stochastic trend	variance shift	combo	PARTIAL	0.003 1	0.73 82
tcpd_seatbelts	192	stochastic trend	mean shift	stationary	mean shift	single_forced_ anom_only_trick	PARTIAL	0.067 8	0.09 27
UNRATE	372	stochastic trend	—	stochastic trend	—	single	FULL	0.032 7	0.10 06
US_investment	104	stationary	—	stationary	—	stat_gate	FULL	0.974 2	0.01 51
uspop	21	deterministic trend	—	stochastic trend	—	single	NONE	0.007 2	0.08 23
W1	45	stationary	point anomaly	stationary	—	stat_gate	PARTIAL	0.390 6	0.97 95
W2	302	stationary	variance shift	stationary	—	stat_gate	PARTIAL	0.181 1	0.95 38
W3	82	stationary	variance shift	stationary	—	stat_gate	PARTIAL	0.138 0	0.98 93
W5	71	deterministic trend	mean shift	stochastic trend	—	single	NONE	0.001 8	0.00 39
W6	114	deterministic trend	variance shift	deterministic trend	variance shift	combo	FULL	0.002 1	0.99 39

Match: yeşil=FULL | sarı=PARTIAL | kırmızı=NONE | Pred Base: yeşil=doğru | kırmızı=yanlış

GT Base Türüne Göre Özet

GT Base	FULL	PARTIAL	NONE	Toplam	FULL%
stochastic trend	4	18	1	23	17%
deterministic trend	1	0	4	5	20%
stationary	6	5	0	11	55%

4. Final Karar Eşikleri ve Blend Formülü

Blend Hesaplama Adımları

Değişken	Hesaplanış	Not
p_single	kazanan_model.predict_proba(X_tsfresh)[:,-1]	Tek anomali modeli — validation F1'e göre seçilen tek model (LGB veya XGB).
p_multi	kazanan_model.predict_proba(X_tsfresh)[:,-1]	Çift anomali modeli — yine tek kazanan model (LGB veya XGB).
meta_X	[P_gate p_base×3 p_single×5 p_multi×5 derived×14 tsfresh×777]	805 boyutlu vektör. Tüm ara model çıktıları + orijinal tsfresh.
p_meta	$0.5 \times \text{XGB_anom}(\text{meta_X}) + 0.5 \times \text{LGB_anom}(\text{meta_X})$	Meta katmanı her zaman XGB+LGB çiftini kaydeder ve blend kullanır.
p_raw	$\alpha \times p_single + (1-\alpha) \times p_multi$	$\alpha=0 \rightarrow$ sadece p_multi; $\alpha=1 \rightarrow$ sadece p_single.
p_final	$0.5 \times p_meta + 0.5 \times p_raw$	Meta ile ham blend eşit ağırlıkla birleştirilir.
Karar	$p_final \geq thr \Rightarrow$ anomali çıktıya eklenir	$thr = thr_anom_only$ (anomaly_only yolu) veya thr_combo (diğer yollar).

Anomali Başına Parametreler

Anomali	Alpha (0=multi, 1=single)	thr_anom_only	thr_combo
collective anomaly	0.0 ($\rightarrow$ pure multi)	0.5	0.7
mean shift	0.0 ($\rightarrow$ pure multi)	0.5	0.8
point anomaly	0.25 ($\rightarrow$ 75% multi / 25% single)	0.4	0.5
trend shift	0.25 ($\rightarrow$ 75% multi / 25% single)	0.4	0.8
variance shift	0.5 ($\rightarrow$ 50% multi / 50% single)	0.3	0.8

Global Eşikler

Parametre	Değer	Ne zaman devreye girer
thr_stat_hard	0.95	$P_gate \geq$ bu değer $\rightarrow$ seri stationary kabul edilir, alt katmanlar atlanır
thr_router	0.3	Router skoru $\geq$ bu değer $\rightarrow$ multi-anom yoluna yönlendirilir